

AI2: Confidence Certificates for Classification

Test set bounds; Occam's razor bounds

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Overview

In this lecture we are interested in the following question:

How reliable is my classifier?

By the end of this lecture, you will learn:

- Test set bounds
- The simplest train-set bound: Occam's razor bound

Learning = Prediction Ability

- Successful learning means that the resulting classifier has prediction ability beyond the examples seen at training.
- We can't expect any prediction ability, in general.
- Here we study prediction ability, assuming an independent samples.

Why Study Prediction Theory?

- 1 To gain insight into how machines' brain works – aka. insights into learning
- 2 Better methods for learning
- 3 Better methods for verifying predictive ability.

Can we verify that learning succeeded?

Standard technique:

- 1 Divide samples into train and test set.
- 2 Train on train set.
- 3 Test on test set.

We can do better! In the sense of a better degree of validity.

Gaining Insight Into Learning

- Overfitting must be avoided by algorithm design
- Example problems in designing learning algorithms:
 - ▶ What is a good pruning criterion?
 - ▶ Why are large margins good?
 - ▶ What other algorithms are likely to yield good results?

Model: Definitions

- $\mathcal{X}$: input space (input domain).
- $\mathcal{Y} = \{0, 1\}$: output space.
- $c : \mathcal{X} \rightarrow \mathcal{Y}$: classifier.

Basic Assumption

All samples are drawn independently and identically distributed (i.i.d.) from some unknown distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$.

- $S = \{(x_i, y_i)\}_{i=1}^n \sim \mathcal{D}^n$: a random i.i.d. sample set.
(training sample or test sample, depending on the context)

Note. The notations we will use in this lecture mostly follow the reference paper given on the last slide to make reading the paper easier.

True Error & Observed Error

A classifier c is good if the following quantity is small:

- **True error:** $c_D \equiv \Pr_{(X,Y) \sim \mathcal{D}}(c(x) \neq y)$.

Problem: This quantity is unknown because the true distribution is unknown.

Instead we have a sample from this unknown distribution. The following quantity is an empirical proxy based on the sample:

- **Observed error:** $\hat{c}_S \equiv \frac{1}{n} \sum_{i=1}^n I[c(x_i) \neq y_i]$

where $I(\cdot)$ denotes indicator function.

Q: Is c_D a deterministic quantity (a number), or a random variable?

True Error & Observed Error

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- **Observed error:** $\hat{c}_S \equiv \frac{1}{n} \sum_{i=1}^n I[c(x_i) \neq y_i]$

where $I(\cdot)$ denotes indicator function.

Q: Is $c_{\mathcal{D}}$ a deterministic quantity (a number), or a random variable?

A: Depends - if c is a fixed function then $c_{\mathcal{D}}$ is deterministic (a number in $[0, 1]$). If c is the classifier trained on the sample S , then $c_{\mathcal{D}}$ is a random variable as a function of c .

Preparation for Test Set Bounds

For the purposes of the Test Set Bound, c is a pre-trained classifier, and S stands for an independent test set.

- **Question:** What is the distribution of $n \cdot \hat{c}_S$?

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- **Question:** What is the distribution of $n \cdot \hat{c}_S$?
- **Answer:** Binomial with mean $n \cdot c_D$.

$$\Pr_{S \sim \mathcal{D}^n}(n \cdot \hat{c}_S = k | c_D) = \binom{n}{k} c_D^k (1 - c_D)^{n-k}$$

This represents the probability of k errors in n flips of a coin with bias parameter c_D .

Binomial PMFs: the distribution of the observed error

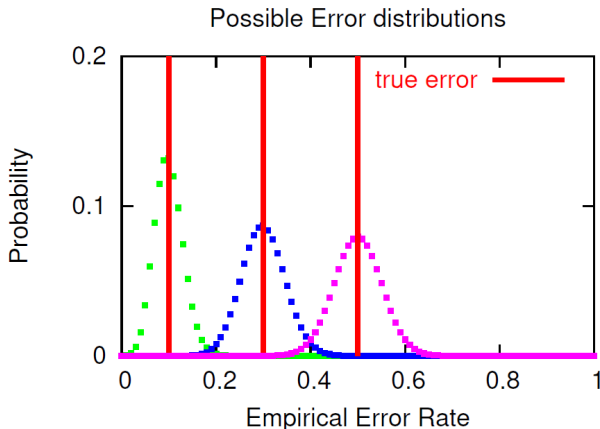


Figure: From: J. Langford: Tutorial on Practical Prediction Theory for Classification, JMLR.

Binomial CDF: the cumulative distribution of the observed error

- **Cumulative distribution:**

$$\begin{aligned}\text{Bin}(n, k, c_{\mathcal{D}}) &= \Pr_{S \sim \mathcal{D}^n}(\hat{c}_S \leq k/n \mid c_{\mathcal{D}}) \\ &= \sum_{i=0}^k \binom{n}{i} c_{\mathcal{D}}^i (1 - c_{\mathcal{D}})^{n-i}\end{aligned}$$

This is the probability of observing at most k (i.e. k or fewer) errors in n trials (test points). Visually, the area under the tail cut of k/n .

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- What we will now do - roughly - is this:
"Plug $n \cdot \hat{c}_S$ for k , set $\text{Bin}(n, n \cdot \hat{c}_S, c_{\mathcal{D}})$ to δ and solve for $c_{\mathcal{D}}$ ",
where $\delta \in [0, 1]$ is a confidence we choose. (To be 95% confident, choose $\delta = 0.05$)

Preparation for Test Set Bound

Choose desired confidence level δ , say 0.1.

- **Pivot of the cumulative:**

$$\overline{\text{Bin}}(n, k, \delta) = \max\{p : \text{Bin}(n, k, p) \geq \delta\}$$

This is the largest true error such that the probability of observing k or fewer "heads" (errors) in n trials (test points) is at least δ .

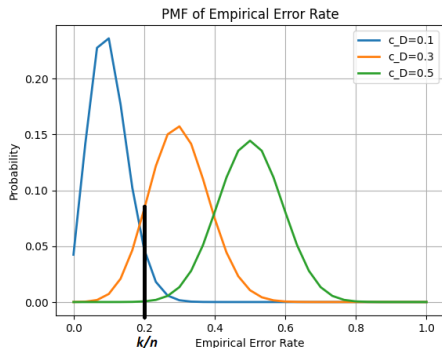
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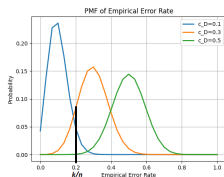
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Visual example: Suppose the observed error rate is $\hat{c}_S = k/n = 0.2$. Then, what is $\overline{\text{Bin}}(n, k, \delta)$ equal to?

(restrict your attention to only the 3 possibilities in the figure)

Worked Exercise for Preparation for Test Set Bound



Visual example: Suppose the observed error rate is $\hat{c}_S = k/n = 0.2$. Then, what is $\overline{\text{Bin}}(n, k, \delta)$ equal to? (restrict your attention to only the 3 possibilities in the figure)

- Recall, the area under each PMF equals 1.
- Recall from definition of CDF that $\text{Bin}(n, k, p)$ is the area of the left tail cut at k/n , of the PMF of the Binomial with true parameter p .
 - For the blue PMF ($p = 0.1$), the tail area of left tail cut at k/n is $> 1/2 \geq \delta$. Pass.
 - For the orange PMF ($p = 0.3$), the area of left tail cut at k/n is $\approx 0.15 - 0.2$ -ish, so $\geq \delta = 0.1$. Pass.
 - For the green PMF ($p = 0.5$), the area of left tail cut at k/n is extremely tiny. If our $\delta = 0.1$ then this would be $< \delta$. No pass.
- Out of the remaining candidates (blue and orange PMFs), we have $\max\{0.1, 0.3\} = 0.3$. Hence, $\overline{\text{Bin}}(n, k, \delta)$ is 0.3.
- In a continuum of candidate PMFs, smaller δ makes $\overline{\text{Bin}}(n, k, \delta)$ larger.

Test Set Bound: Theorem

Theorem (Test Set Bound)

For all classifiers c , for all distributions $\mathcal{D}$, for all $\delta \in (0, 1]$:

$$\Pr_{S \sim \mathcal{D}^n} (c_{\mathcal{D}} \leq \overline{\text{Bin}}(n, n \cdot \hat{c}_S, \delta)) \geq 1 - \delta$$

- This result provides a probabilistic guarantee on the true error rate based on observed data.
- It is derived under the assumption of i.i.d. samples from the distribution $\mathcal{D}$.

Proof.

The probability that $\hat{c}_S$ will not fall into a tail of size δ is exactly $1 - \delta$. Hence, with this probability, $\text{Bin}(n, n \cdot \hat{c}_S, c_{\mathcal{D}}) \geq \delta$, so by definition of $\overline{\text{Bin}}$, we have $c_{\mathcal{D}} \leq \overline{\text{Bin}}(n, n \cdot \hat{c}_S, \delta)$. □

How to use the test set bound?

- For the tightest bound, we can use numerical methods to compute $\overline{\text{Bin}}(n, n \cdot \hat{c}_S, \delta)$ – example code provided in Canvas.
- For reasons of simplicity, we can use upper-bounds on $\overline{\text{Bin}}(n, n \cdot \hat{c}_S, \delta)$ instead at the expense of giving up some tightness (as we shall see shortly).

Simplified forms of the bound

There are upper bound approximations on $\overline{\text{Bin}}(n, k, p)$ that are simpler to compute and simpler to interpret (at the expense of the bound becoming slightly less tight). One of these result in the following corollary.

Corollary. For all classifiers c , for all distributions $\mathcal{D}$, for all $\delta \in (0, 1]$:

$$\Pr_{S \sim \mathcal{D}^n} \left(c_{\mathcal{D}} \leq \hat{c}_S + \sqrt{\frac{\ln \frac{1}{\delta}}{2n}} \right) \geq 1 - \delta.$$

This corollary is also obtainable from Hoeffding's bound - which you will encounter again in the ML module.

Remember that it can be used on its own to certify learning only if S was the independent test set (not the training set!) – think about why!

What Does the Test Set Bound Tell Us?

- The test set bound gives a mechanism to certify the extent to which the classifier has learned to predict correctly.
- The guarantee is probabilistic - it only implies that the bound will not be wrong for a $1 - \delta$ fraction of its applications / i.e. training sets.
- The bound tightens with larger sample sizes n .
- Relies on the assumption of independent and identically distributed (i.i.d.) samples.
- Useful for both theoretical insights and practical applications.
- Simple and its validity is well understood (as opposed to empirical method e.g. cross-validation type of methods)

Worked Example: Using the Test Set Bound

- Suppose $n = 100$ samples.
- Observed error $\hat{c}_S = 0.1$ (10 errors on the test set).
- Use the Corollary to the Test Set Bound to give an upper bound on the true error with 95% confidence.

Applying the Bound

Set confidence level $1 - \delta = 0.95$ ($\delta = 0.05$).

$$c_{\mathcal{D}} \leq \hat{c}_S + \sqrt{\frac{\ln \frac{1}{\delta}}{2n}}$$

Substitute values:

$$c_{\mathcal{D}} \leq 0.1 + \sqrt{\frac{\ln 20}{200}} \approx 0.1 + 0.08 = 0.18$$

- Interpretation: With 95% confidence, the true error rate $c_{\mathcal{D}}$ is at most 18%.

A Brief Intro to Training Set Bounds

Question: Is it possible to derive a similar learning guarantee if S is the training sample?

Why this question is interesting:

- We might not have enough labelled data to afford an independent test set
- Gain more insights into learning
- Inspire new algorithms

Occam's razor bound is the simplest train-set bound.

For the purpose of train-set bounds, S is the training sample, and there is no test sample.

A Brief Intro to Training Set Bounds

First let's understand why the Test Set Bound we already got is no longer valid if S is the training set.

- Recall, the test set bound was derived under the assumption that the coin-flips (errors $I[c(x_i)) \neq y_i]$) are i.i.d.
- If S was an independent test set then this is true provided the examples are i.i.d.
- If S the training sample, then the classifier c becomes a random variable as it depends on this S , hence all $I[c(x_i)) \neq y_i]$ depend on the same c – they are no longer i.i.d!
- Therefore, the same bound would not hold true when S is the training set!
- Next, we extend the derivation to work when S is the training sample.

Occam's Razor Bound - Preparations

- Confidence bounds are frequentist guarantees – including both Test Set Bounds, and Train Set Bounds.
- However, we will take and adapt a Bayesian idea.
- Before seeing the training set S , we define a "prior" probability distribution over the classifiers, $P(c)$. This represents our "bet" on the candidate classifiers.
- Suppose a simple learning problem where we have a countable number of classifiers for the learning algorithm to choose from based on a training sample S .

Occam's Razor Bound

Theorem (Occam's Razor Bound)

For all "priors" $P(c)$ over a countable set of classifiers, for all distributions $\mathcal{D}$, for all $\delta \in (0, 1]$:

$$\Pr_{S \sim \mathcal{D}^n} (\forall c : c_{\mathcal{D}} \leq \overline{\text{Bin}}(n, n\hat{c}_S, \delta P(c))) \geq 1 - \delta$$

Compare with Test Set Bound: $\delta \rightarrow \delta P(c)$.

Corollary. For all $P(c)$, all $\mathcal{D}$, all $\delta \in (0, 1]$:

$$\Pr_{S \sim \mathcal{D}^n} \left(c_{\mathcal{D}} \leq \hat{c}_S + \sqrt{\frac{\log \frac{1}{P(c)} + \log \frac{1}{\delta}}{2n}} \right) \geq 1 - \delta.$$

Tightness depends on the self-information $\log \frac{1}{P(c)}$ of the classifier returned by the learning algorithm, with respect to our prior bets.

Occam's Razor Bound: Proof

Idea: Without training, apply the Test Set Bound to each c in turn with confidence parameter $\delta P(c)$:

$$\forall c \quad \Pr_{S \sim D^m} (c_D \leq \overline{\text{Bin}}(m, n\hat{c}_S, \delta P(c))) \geq 1 - \delta P(c)$$

Negate to get equivalent statement:

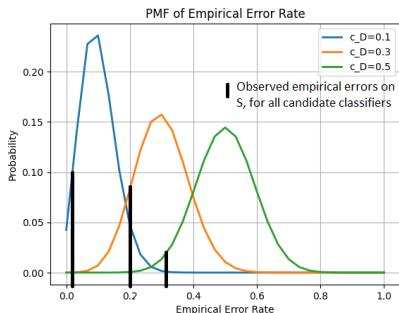
$$\forall c \quad \Pr_{S \sim D^m} (c_D > \overline{\text{Bin}}(m, n\hat{c}_S, \delta P(c))) < \delta P(c)$$

Apply **union bound**: $\Pr(A \text{ or } B) \leq \Pr(A) + \Pr(B)$ repeatedly.

$$\Pr_{S \sim D^m} (\exists c : c_D > \overline{\text{Bin}}(m, n\hat{c}_S, \delta P(c))) < \sum_c \delta P(c) = \delta$$

Negate again to complete the proof. $\square$

Illustration and Discussion



- Each classifier's error distribution is a Binomial; each have their own true error and observed error.
 - The union bound makes sure that none of the observed errors falls into the tail of size $\delta P(c)$ of their respective candidate classifier c .
 - The training algorithm will choose one of them.
-
- Size of tail $\delta P(c)$ is proportional to our bet – so good bet gives tight bound, poor bet gives loose bound, but the bound is always valid given its assumptions.
 - The assumptions in Occam's bound: i.i.d. sample, and countably many candidate classifiers.

Conclusions

- The Test Set Bound is a high-probability $(1 - \delta)$ confidence bound on classifiers' unknown true error rate, computed using an independent test set.
- Very simple and tight.
- (There is also a similarly tight lower bound, giving us a real confidence interval on classifier predictions.)
- The Occam's Razor bound is the simplest training-set bound – giving a confidence bound on classifiers' unknown true error rate without access to an independent test set.
- The tightness of the Occam's Razor Bound depends on the self-information (according to our prior bet) of the classifier returned by the training algorithm.

Reference:

John Langford: Tutorial on Practical Prediction Theory for Classification, JMLR

If you like C++, have a look here: https://hunch.net/~jl/projects/prediction_bounds/bound/bound.html